

Project Euler

Ihsaan Abrar Hidayathullah Baig

Problem 50: Consecutive Prime Sum

Consecutive Prime Sum

Problem 50



The prime 41, can be written as the sum of six consecutive primes:

$$41 = 2 + 3 + 5 + 7 + 11 + 13.$$

This is the longest sum of consecutive primes that adds to a prime below one-hundred.

The longest sum of consecutive primes below one-thousand that adds to a prime, contains 21 terms, and is equal to 953.

Which prime, below one-million, can be written as the sum of the most consecutive primes?

Answer:

Check

The code finds the prime number below one million that can be formed by adding the greatest number of consecutive prime numbers. It checks the longest possible sequences first and outputs.

```
def is_prime(number):
    if number < 2:
        return False

    divisor = 2

    while divisor * divisor <= number:
        if number % divisor == 0:
            return False

        divisor += 1

    return True

def solve(limit):
    primes = []

    for number in range(2, limit):
        if is_prime(number):
            primes.append(number)

    length = 1

    while sum(primes[:length]) < limit:
        length += 1

    for length in range(length - 1, 0, -1):
        for start in range(len(primes) - length + 1):
            total = sum(primes[start:start + length])

            if total >= limit:
                break

            if is_prime(total):
                return total

print(solve(1_000_000))
```

997651



Congratulations, the answer you gave to problem 50 is correct.